SMART HEALTHCARE ANALYTICS

Tableau Dashboard and Story — Project Analysis Report

Prepared by: Mustafa Bartanwala

Report Purpose

This report documents the structure and observed visual analyses in the Smart Healthcare Analytics Tableau project. It summarizes the dashboard objectives, charts, visible patterns, and limitations. Numerical observations are included only where they were available from the project screenshots and notes.

Project report | October 2026

1. Project Objective

The objective is to explore a healthcare dataset in Tableau and present patient demographics and selected health indicators through worksheets, a dashboard, and a Tableau Story. The analysis focuses on age, BMI, glucose, cholesterol, heart disease, smoking status, alcohol status, and chest pain.

2. Tools Used

• Tableau Public / Tableau Desktop — worksheets, visualizations, dashboard, and story.
• CSV healthcare dataset — source data for the visualizations.
• GitHub — project files and documentation.

3. Dataset Overview

The project dataset contains 5,000 records. Fields described for the analysis include age, gender, BMI, blood pressure, cholesterol, glucose, smoking, alcohol consumption, exercise level, fatigue, dizziness, chest pain, heart disease, diabetes, stroke, and health_risk_score. The available project notes report no missing values or exact duplicate rows. Age ranges from 18 to 79, and BMI ranges from 15 to 42.6.

The health_risk_score field was reported as constant at 100, so it is not useful for comparing risk across records.

4. Visualizations and Analysis

Visualization	Purpose and observed interpretation
Age Group Distribution	Counts patients across age groups 18–29, 30–44, 45–59, and 60+. In the reviewed chart, the 60+ group appeared largest and the 18–29 group smallest.
BMI Category Distribution	Displayed counts were approximately 516 underweight, 2,010 normal range, 1,671 overweight, and 803 obesity range. Normal-range BMI was the largest category.
Heart Disease by Age Group	Uses the average of a 0/1 heart-disease indicator, formatted as a percentage. The chart showed an unusually high value for age 60+ (about 87.6%) and some zero values. Validate these against the source data before drawing conclusions.
Glucose by Diabetes Status	A box-and-whisker plot compares glucose distributions for diabetes versus no recorded diabetes. It supports comparison of median, spread, and outliers; exact statistics were not recorded.
Cholesterol by Heart Disease Status	A box-and-whisker plot compares cholesterol distributions by heart-disease status. Exact medians and spreads were not captured in the available notes.
Heart Disease by Smoking Status	The chart showed about 28.72% for non-smokers and 31.21% for smokers. This is a descriptive association, not proof of causation.
Heart Disease by Alcohol Status	The chart showed about 30.32% for alcohol consumers and 29.64% for non-consumers. The displayed difference is small and does not establish causation.
Chest Pain Analysis	A pie chart summarizes chest-pain status categories. Exact counts or percentages were not recorded in the available notes.
Risk Factor Heatmap	Uses age group and smoking status as dimensions and heart-disease prevalence as color/label to compare combinations. Check for empty or sparse groups.

5. Dashboard and Tableau Story

The project brings nine visualizations into one dashboard and uses a Tableau Story to present analyses in sequence. The story includes separate points for age distribution, BMI, heart disease by age, glucose, cholesterol, smoking, alcohol, chest pain, and the risk-factor heatmap.

6. Key Observations

• Normal-range BMI was the largest BMI category in the displayed chart, followed by overweight, obesity range, and underweight.
• The smoking comparison displayed higher heart-disease prevalence for smokers than non-smokers (about 31.21% versus 28.72%).
• The alcohol comparison showed a small difference (about 30.32% versus 29.64%).
• The age-based heart-disease chart contains values that should be validated before reporting age-related conclusions.
• The glucose and cholesterol box plots allow distribution comparisons, but exact statistics were not captured.

7. Limitations

• This report uses chart descriptions, project notes, and values visible in reviewed screenshots; it is not a fresh statistical computation from the raw CSV.
• The unusually high age 60+ prevalence and zero values in the age chart should be checked for category assignments, encoding, and aggregation.
• health_risk_score was reported as constant at 100 and cannot distinguish risk between records.
• Observational differences do not establish causation and may reflect confounding factors.
• Dataset representativeness, collection method, and clinical validity were not independently established.

8. Recommendations

1. Verify heart_disease is consistently encoded as 0 and 1 and that the calculated indicator maps it correctly.
2. Check Age Group assignments and confirm that each group contains the expected records.
3. Recheck filters and aggregation for the age-based chart and heatmap.
4. Add the Tableau Public link to the README after publishing.
5. Describe results as exploratory patterns; avoid clinical or causal claims.

9. Conclusion

The Smart Healthcare Analytics project organizes healthcare data into nine Tableau visualizations and a dashboard/story experience. The charts offer ways to explore patient age, BMI, glucose, cholesterol, chest pain, and displayed associations between heart disease and smoking or alcohol status. The age-based prevalence values should be validated against the raw data before final submission.

Source Note

Prepared from the healthcare project details and chart observations shared during the project workflow. Approximate values were transcribed from reviewed chart screenshots/notes and should be confirmed in Tableau before use in formal conclusions.